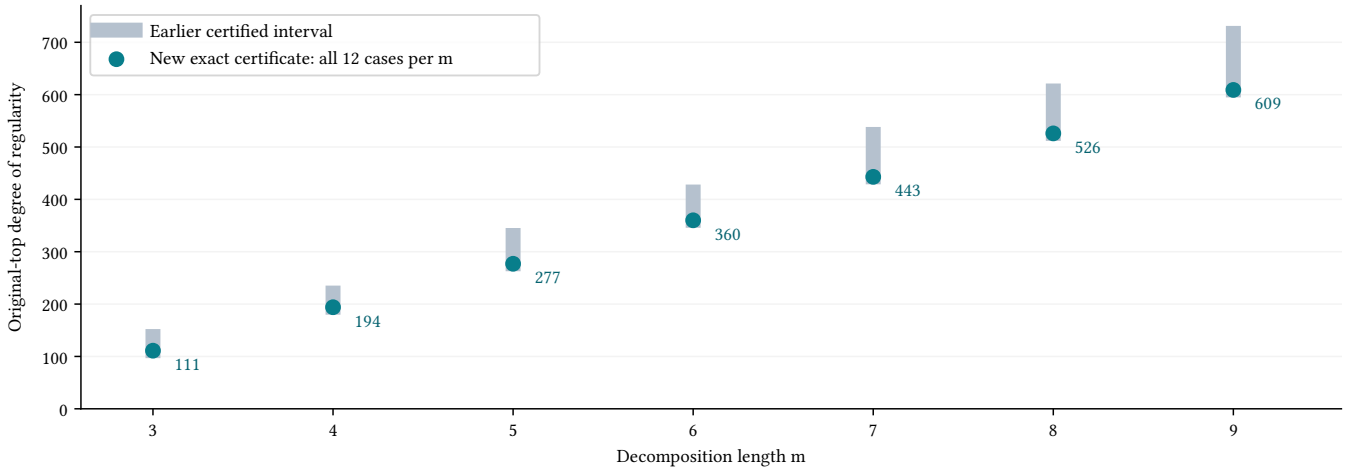
Exact original-top binary chain degrees

84 saved systems · 84 fresh replays · 84 S3-restored replays

Matching lower and upper certificates determine the original-top degrees for all retained binary/Koblitz systems at m=3–9. Native checks verify the hypotheses of the written overlap proof. **Full algebraic Lean formalization and independent scientific review remain pending.** These are not measured solver degrees.

m	Earlier interval	Exact original-top degree	Cases
3	111–138	**111**	12
4	194–221	**194**	12
5	277–331	**277**	12
6	360–414	**360**	12
7	443–524	**443**	12
8	526–607	**526**	12
9	609–717	**609**	12



Run: n83-chain-exact-20261029-r1. Recorded degree certificates, twelve retained cases per m.

Each base uses 2^27 candidate x-values per curve. Each row covers original, baseline and spike40, both binary families and both target kinds. Bases remain fixed across m=2–9; no points, targets or systems changed. Natural repeats are provenance identities, not independent samples.

Context. The prior m=2 degrees remain Koblitz 28, binary 28 and prime 42. Prime m=3–9 original top quotients remain positive-dimensional, with no finite first vanishing degree. The48-row fixed-base view now has34 exact and14 prime entries. All earlier intervals and views are preserved. This is improved knowledge of unchanged systems; it is not an elite-base gain or an FFD/LFD conclusion.

Proof structure, controls and preserved evidence

1. Verify the original presentation. Bind every input and actual point-base manifest by hash. Reconstruct all original highest coefficients and complete supports; check ring, field equations and exact witness suffix. No affine consequence is substituted before taking highest forms.

2. Account for overlapping components. Full coefficient Frobenius diagonalizes all field-product cubics. All 83 cyclic27-row selector windows have verified inverse pairs. Counting window intersections gives a rank lower bound for every embedding subset. Decomposition by intermediate-variable monomial support then bounds the augmented product quotient by socle $83(m-2)+27$.

3. Preserve the terminal defect. For nonzero target, the terminal quadratic leaves a possible product direction h in degree 2. All terminal local variables annihilate h. A truncated-chain quotient bounds its shifted module by $83(m-3)+56$, which is below the augmented bound. The original-source projection retains $83(m-2)+27$ variables, giving equality and first vanishing degree one higher. Zero target is checked separately through invertible coefficient Frobenius.

Controls. Five native tests pass, including 44 complete small mixed-degree Hilbert calculations. All 36 cases inside the hypotheses match. All 8 negative cases give degree 7 rather than formula 6 and are refused by the guard. Saved augmented/truncated Hilbert arrays, kernel differences and exhaustive support checks are independent of the large-case bound. Two Lean lemmas prove only the arithmetic implications, using standard propext and Quot.sound. Full polynomial, rank-decomposition and graded-module bridges remain written, not formalized.

Preservation and replay. All 360 raw manifest entries matched before and after downloaded-data replay. All 84 immutable cumulative progress snapshots, original inputs, executable, protocol, controls, certificates and errors remain. Data were committed and pushed before this report. Read CHAIN_EXACT_FORMAT.json, CHAIN_EXACT_DESIGN.md, the unified export and evidence/chain-exact-storage.json. Producer replay does not substitute for independent scientific review.

Archive SHA256: `a0f135c6a765d04f9169f875cd90a73b56c97db42f4c99bb9595949a0925ced3`
Source: `15b67074130e90f92852b261f46e253b74eab338`
Protocol: `35d448ff9521388239b6aaca3cb6efaeca9a354b0d32d7d97a7ccf44fffd5c29`
Data commit: `7ea0bcc977e315edadb1af6c5d2b384fdec5dc48`

The archive is in setec-astronomy-artifacts under store/sha256, first two hash characters and full hash. Extract fresh, check manifest, replay, recheck manifest. Existing-host fallback followed a failed EKS storage probe with unconfirmed cause; refreshed logs were empty. No timing, speed or attack claim.

Remaining work. FFD/LFD, prime-case completion, independent review and held-out elite-base evidence. The tested linear selector bases have equal original-top degrees. A basis change alone in this theorem's class cannot improve this metric; other invariants or presentations require their own explicit protocols.